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Teaching machine data and Excel to speak one language, in real time

A leading mining organisation in the Middle East engaged Sthiros to digitise plant reporting, replacing Excel with real-time Power Apps and BI dashboards across every plant function.

Client	Service	Sector	Geography
Leading Mining Organisation	**Plant Digitisation & Integration, Power Apps, Data Integration & BI Reporting**	**Mining, Plant Operations**	**Middle East**

The Challenge:

When machines and people don't share a data model

This mining organisation ran plant reporting on Excel, with no standardised capture across plants and no link between PLC/SCADA machine data and manual inputs. Performance issues surfaced only after delays in consolidation, never in real time.

Excel-Dependent Reporting

Manual entry and reporting on Excel caused delays, errors, and inconsistencies.

No Standardised Capture

No common system existed for capturing operational data across plants.

OT and IT Disconnect

Plant systems and enterprise reporting ran as two separate, unconnected worlds.

No Real-Time Visibility

No centralised dashboards existed to monitor plant performance as it happened.

Consolidation Delays

Manual data consolidation across Excel files slowed every reporting cycle.

Machine-Human Data Gap

Machine tag data and manual inputs had no shared model to combine them.

The Solution:

One data model, replacing two languages with one

Sthiros replaced Excel with structured Power Apps forms, then integrated PLC and SCADA machine data with manual inputs into one centralised data model.

Power BI dashboards now deliver real-time KPIs and trend analysis in place of static reports, across every plant function.

The Methodology

1

Digitisation Layer, Power Apps

Excel-based processes replaced with mobile-friendly, validated Power Apps forms, standardised across every plant.

2

Data Integration Layer, OT + IT Convergence

PLC and SCADA tag data integrated via OPC UA, combined with manual inputs into one centralised SQL model.

3

Data Architecture & Infrastructure

A hybrid OT, IT, and cloud architecture built with secure gateways and auto-sync recovery for resilience.

4

Analytics & Reporting Layer

Power BI dashboards deliver real-time KPIs, trend analysis, and centralised reporting across desktop and mobile.

What This Means

OT and IT, one number.

Slash manual entry by 70%.

Same setup, added plants.

Outcomes

100%

Excel reporting eliminated

Core manual reporting processes fully replaced with digital Power Apps capture.

50-70%

Manual entry effort cut

Structured, validated data capture reduced rework and reporting turnaround time.

3+

Core functions digitised

Plant Operations, Engineering, and Management now run on one unified platform.

Near Real-Time

Data availability

Machine and manual data fused into a single, centralised SQL-based model.

Get in touch

Is your plant data connected, or just collected?

Speak to Sthiros about what a structured digitisation programme can unlock for your plant operations.

Connect with us:
sthiros.com

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